# Lattice — Feature Showcase

> **Open this file with Lattice** to see every feature rendered live.
> Switch between **Edit**, **Preview**, and **Dual** view with the toolbar selector.
> Try `Ctrl/Cmd+Shift+T` to auto-generate the Table of Contents below.

## Table of Contents

- Keyboard Shortcuts Reference

## Text Formatting

Normal paragraph text flows here. Lattice renders Markdown as **close to GitHub** as possible, so files look identical whether viewed in the editor preview, on GitHub, or printed.

| Style | Syntax | Result |
|---|---|---|
| Bold | `**bold**` | **bold** |
| Italic | `*italic*` | *italic* |
| Bold + italic | `***both***` | ***both*** |
| Strikethrough | `~~strikethrough~~` | ~~strikethrough~~ |
| Inline code | `` `code` `` | `code` |
| Superscript | `X<sup>2</sup>` | $X^2$ |
| Subscript | `H<sub>2</sub>O` | $H_2O$ |

> **Note — sub/superscript:** CommonMark/GFM has no native sub/superscript syntax. Lattice renders `<sup>` and `<sub>` HTML tags directly (GitHub and Obsidian do too). We recommend, though that you use alternatives that work everywhere: KaTeX math ( `$X^2$` → $X^2$, `$H_20$` → $H_2O$) or Unicode characters ( $X^2$ , $H_2O$ ). The linter flags these tags as a portability hint — they may not render in all renderers.

## Highlight Marks

Highlight marks are a Lattice-exclusive feature. Wrap any text in `==double equals==` to render it with a bright amber background.

This works in **any context** — inside a sentence, ==across *italic* and **bold** spans==, or on its own line:

This entire sentence is highlighted.

> [!NOTE] Highlight marks are **not** rendered inside code blocks or inline code — `==this stays raw==` .

## Lists

### Unordered

- First item
- Second item
  - Nested item A
  - Nested item B
    - Deeply nested
- Third item with **bold** and *italic* text

### Ordered

1. Install Lattice from the [releases page]
2. Open any `.md` file — double-click or via **File → Open**
3. Choose your view:
   i. **Edit** — raw CodeMirror editor only
   ii. **Preview** — rendered HTML only
   iii. **Dual** — side-by-side (four layout variants)
4. Print with `Ctrl/Cmd+P` — always renders to a clean light theme

### Definition-style (using bold + indent)

**Local-first** All files stay on your machine. No cloud sync required.

**Documents as Code** Plain-text `.md` files are diff-friendly, version-controlled, and AI-ready.

## Task Lists (GFM)

GitHub Flavored Markdown task lists render as interactive checkboxes in Lattice.

- [x] Install Lattice
- [x] Open the demo file
- [x] Explore Edit / Preview / Dual views
- [ ] Try `Ctrl/Cmd+Shift+T` to regenerate the Table of Contents
- [ ] Try `Ctrl/Cmd+Shift+L` to auto-align a pipe table

- [ ] Paste a screenshot with `Ctrl/Cmd+V` — it auto-saves to a sibling folder
- [x] Open a second file in a new window (multi-window support)
- [ ] Try printing to PDF with `Ctrl/Cmd+P`

## Tables

Lattice supports **GFM pipe tables** with left, center, and right alignment. Press `Ctrl/Cmd+Shift+L` with the cursor anywhere in the table to auto-format it.

| Feature | Edit View | Preview View | Dual View |
| --- | :---: | :---: | :---: |
| CodeMirror editor | ✓ | — | ✓ |
| Live rendered preview | — | ✓ | ✓ |
| Scroll sync | — | — | ✓ |
| Print (Ctrl+P) | ✓ | ✓ | ✓ |
| ==Highlight marks== | ✓ | ✓ | ✓ |
| Mermaid diagrams | — | ✓ | ✓ |
| KaTeX mathematics | — | ✓ | ✓ |

## Code

### Inline code

Use backticks for `inline code`. Great for `file/paths`, `commands`, and `variable_names`.

### Fenced code blocks

Lattice passes code fences to the preview unchanged — syntax highlighting is provided by the renderer.

```
// Lattice highlight-mark plugin — converts ==text== to <mark> elements
export const MARK_RE = /==((?:[^=\n]|=(?!=))+?)==/g;

export function splitAtMarks(text: string): HastNode[] {
```

```
  const parts: HastNode[] = [];
  let lastIndex = 0;
  MARK_RE.lastIndex = 0;
  let match: RegExpExecArray | null;
  while ((match = MARK_RE.exec(text)) !== null) {
    if (match.index > lastIndex)
      parts.push({ type: 'text', value: text.slice(lastIndex, match.index) });
    parts.push({
      type: 'element', tagName: 'mark',
      properties: {},
      children: [{ type: 'text', value: match[1] }],
    });
    lastIndex = match.index + match[0].length;
  }
  if (lastIndex < text.length)
    parts.push({ type: 'text', value: text.slice(lastIndex) });
  return parts;
}
```

```
// Tauri command — read file with conflict detection
#[tauri::command]
pub async fn read_file(path: String) -> Result<FileReadResult, String> {
    let content = fs::read_to_string(&path)
        .map_err(|e| format!("Cannot read {path}: {e}"))?;
    let hash = compute_sha256(&content);
    Ok(FileReadResult { content, hash })
}
```

```
# Build a production release
export LATTICEBUILD_NO="MyBuild-001"
npm run tauri build
```

## Blockquotes

> **Single-level blockquote.** Great for callouts, citations, or side notes.

> Nested blockquotes are also supported:
>
> > Inner quote — "*Documents as code, not documents as data.*"
> >
> > — Lattice design philosophy

### GitHub-style Alerts (rendered by GitHub and Lattice)

> [!NOTE] Useful information that users should know, even when skimming.

> [!TIP] Optional advice for a better experience.

> [!IMPORTANT] Key information users need to know to achieve their goal.

> [!WARNING] Urgent information that needs immediate attention.

> [!CAUTION] Advises about risks or negative outcomes.

## Mathematics — KaTeX

Lattice renders mathematics via **KaTeX**. Use `$...$` for inline and `$$...$$` for display math.

### Inline math

Einstein's famous equation $E = mc^2$ and the quadratic formula $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ render inline.

The Euler identity $e^{i\pi} + 1 = 0$ is often called the most beautiful equation in mathematics.

### Display math — Calculus

$$\int_{-\infty}^{\infty} e^{-x^2}\, dx = \sqrt{\pi}$$

$$\frac{d}{dx}\left[\int_a^{g(x)} f(t)\, dt\right] = f(g(x)) \cdot g'(x)$$

### Display math — Linear Algebra

$$\mathbf{A}\mathbf{x} = \lambda\mathbf{x} \quad \Longrightarrow \quad \det(\mathbf{A} - \lambda\mathbf{I}) = 0$$

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad - bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

### Display math — Series & Limits

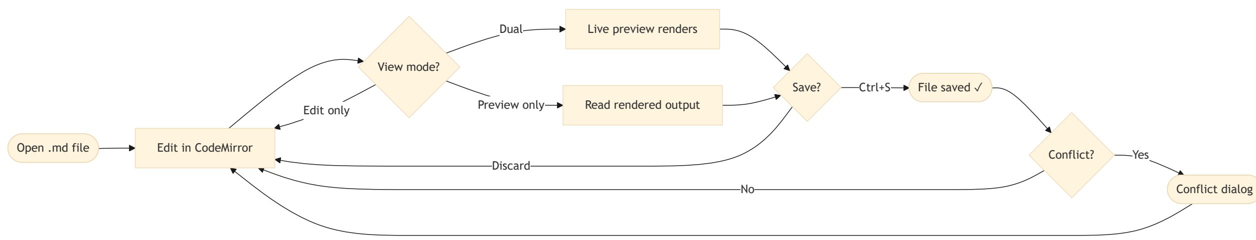
$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots$$

$$\lim_{n \to \infty} \left(1 + \frac{1}{n}\right)^n = e$$

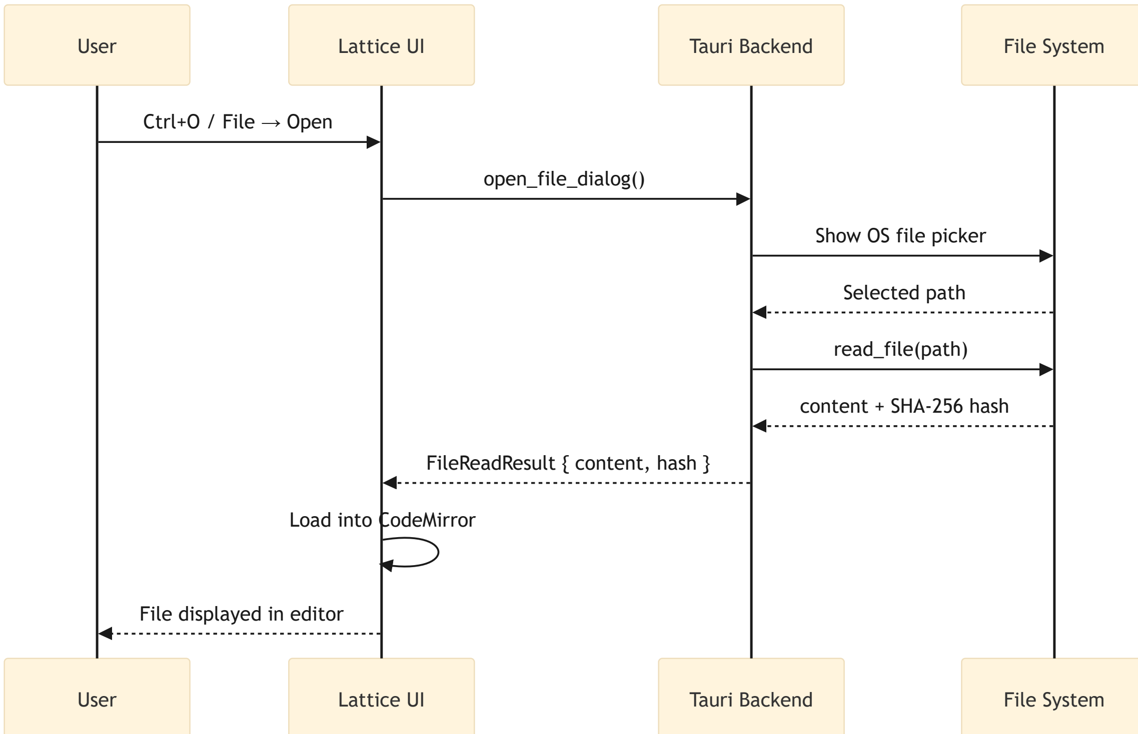
## Mermaid Diagrams

Lattice renders ` ```mermaid ``` ` fenced blocks inline in the preview.

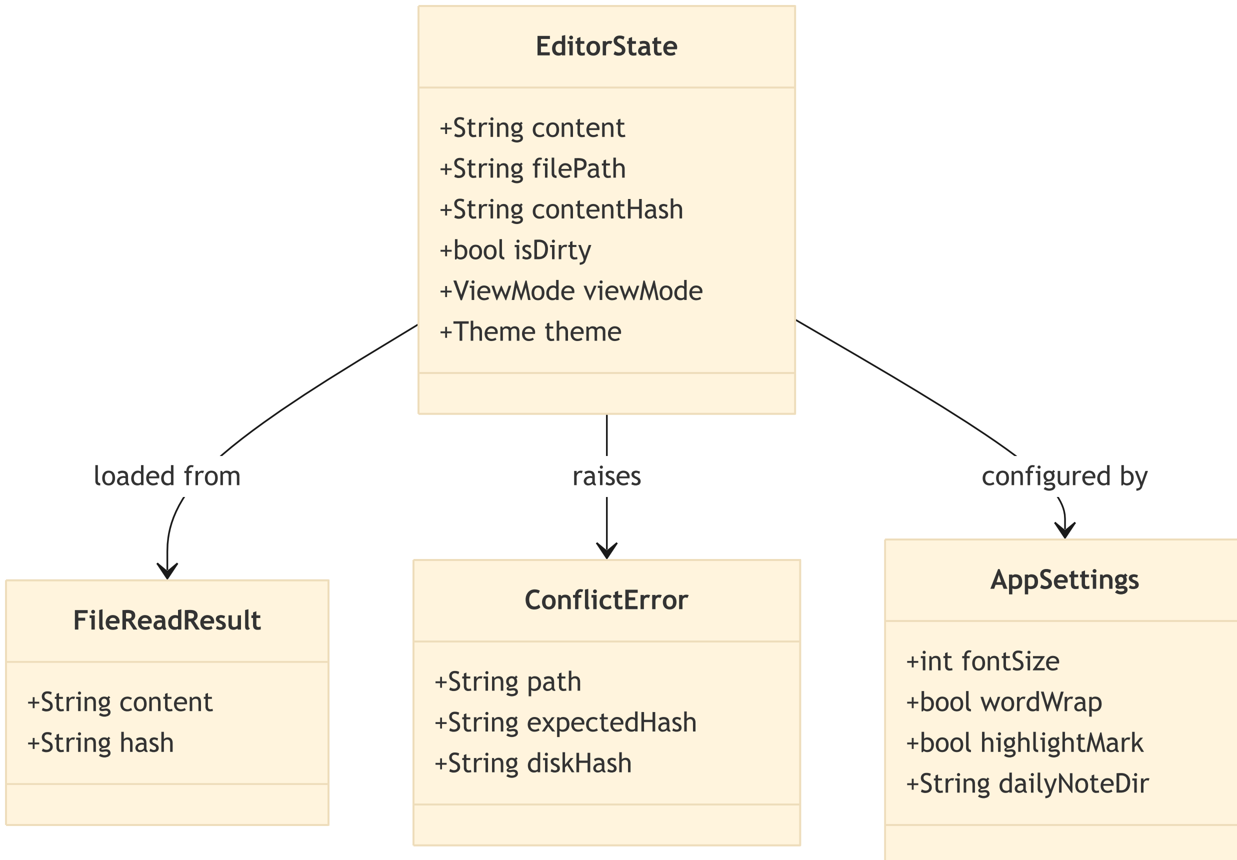
### Flowchart — Lattice Edit Cycle



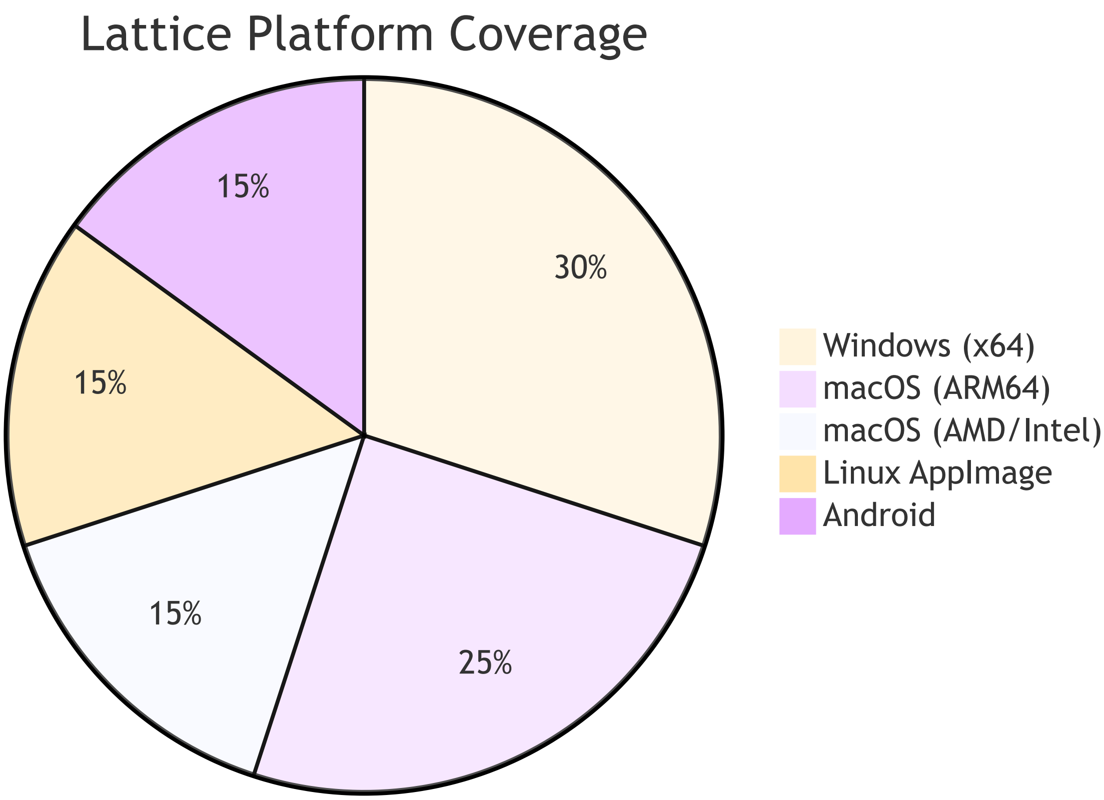

### Sequence Diagram — File Open Flow




## Class Diagram — Core Data Model




## Pie Chart — Supported Platforms



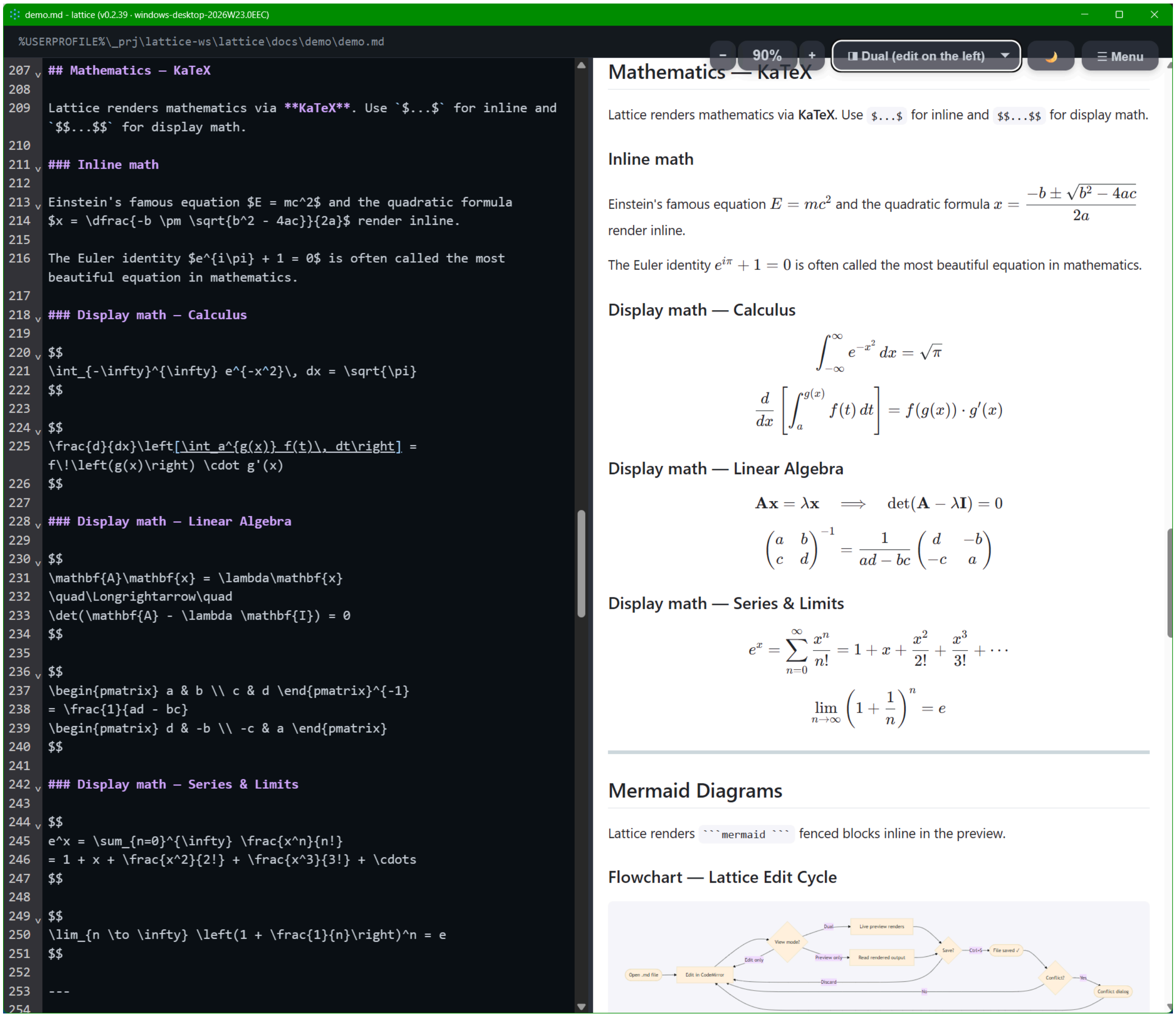

## Images

Paste images directly into Lattice with `Ctrl/Cmd+V` .
The image file is automatically saved to a sibling folder named after the current document, and a Markdown reference is inserted at the cursor position:

```
![Image](demo_assets/img_1780574419628.png)

![Image](demo_assets/img_1780575215039.png)
```

**Here is how it renders in `Dual (edit on the left)` mode**:




Here is how it renders in `Dual (edit on the bottom)` mode:

demo.md - lattice (v0.2.39 · windows-desktop-2026W23.0EEC)

%USERPROFILE%_prj\lattice-ws\lattice\docs\demo\demo.md

– 70% + Dual (edit on bottom) ☰ Menu

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$$

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```
---

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```

```
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```

> **Tip:** The assets folder name is derived from the document filename, keeping related files grouped together and easy to version-control.

## Links

- Lattice on GitHub — source code and issue tracker
- Releases Page — download installers
- Release Site — GitHub Pages release portal
- Relative link: System Architecture Demo
- Relative link: Physics Equations Demo

Auto-link: https://github.com/blessia-blessini/lattice

## HTML in Markdown

Lattice renders a **safe whitelist** of inline HTML tags and passes everything else through as literal text. The linter flags all inline HTML — whitelisted tags get a **hint** (portability risk), non-whitelisted tags get a **warning** (not rendered here).

| HTML element | Lattice preview | GitHub | Portable alternative |
|---|---|---|---|
| <sup> | ✅ rendered | ✅ | KaTeX: `$X^2$` → $X^2$ · Unicode: `X²` |
| <sub> | ✅ rendered | ✅ | KaTeX: `$H_2O$` → $H_2O$ · Unicode: `H₂O` |
| <kbd> | ✅ rendered | ✅ | Inline code: `` `Ctrl+S` `` |
| <br> | ✅ rendered | ✅ | Two trailing spaces or blank line |
| <mark> | ⚠️ plain text | ✅ | `==highlight==` (Lattice-native) |
| <details> | ⚠️ plain text | ✅ | Headings + TOC (no universal equivalent) |
| Other HTML | ⚠️ plain text | varies | Markdown equivalent (case-by-case) |

Live examples of the whitelisted tags:

- Superscript: $X^2$ + $Y^3$
- Subscript: $H_2O$ and $CO_2$

- Keyboard: press `Ctrl` + `S` to save

## Keyboard Shortcuts Reference

| Action | Windows / Linux | macOS |
| --- | --- | --- |
| Save | `Ctrl+S` | `Cmd+S` |
| Open file | `Ctrl+O` | `Cmd+O` |
| New window | `Ctrl+N` | `Cmd+N` |
| Generate / refresh TOC | `Ctrl+Shift+T` | `Cmd+Shift+T` |
| Auto-align table | `Ctrl+Shift+L` | `Cmd+Shift+L` |
| Paste image from clipboard | `Ctrl+V` | `Cmd+V` |
| Print / Save as PDF | `Ctrl+P` | `Cmd+P` |
| Increase font size | `Ctrl++` | `Cmd++` |
| Decrease font size | `Ctrl+-` | `Cmd+-` |